Alejandro Scaffa, PhD

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Profile

PhD-trained biomedical scientist with a doctorate in Molecular Pharmacology and Physiology from Brown University and postdoctoral research experience at Merck Research Laboratories. Doctoral research examined the long-term impact of supplemental oxygen exposure on premature lung development, with a focus on cellular senescence and metabolic dysfunction. Postdoctoral work at Merck focused on cancer biology, the tumor microenvironment, and senescence, including analysis and interpretation of single-cell RNA-sequencing data. Currently completing an M.S. in Computer Science, with training in algorithms, machine learning, and reproducible data analysis, and focusing on careers at the interface of experimental biology, computation, and pedagogy.

Education

Northeastern University (Roux Institute), Portland, Maine, USA
M.S. Computer Science (GPA 3.94) — Expected Aug 2026

Brown University, Providence, Rhode Island, USA
Ph.D. Molecular Pharmacology and Physiology — May 2020
Mentor: Phyllis Dennery M.D.
M.A. Molecular Pharmacology and Physiology — May 2016

Grinnell College, Grinnell, Iowa, USA — May 2014
B.A. with honors, Biochemistry

PhD Research

Hyperoxia (Supplemental oxygen) leads to Senescence and Glycolytic Dysregulation in ATII lung cells, a model for Bronchopulmonary Dysplasia - P.I.: Prof. Dennery MD
PhD focused on negative sequelae from life-saving lung ventilation with supplemental oxygen (hyperoxia) given to premature babies. Described mechanisms leading to hyperoxia-induced senescence in vitro and in vivo, and compositional changes to the lung after hyperoxia through scRNAseq and innovative imaging analysis.

Professional Experience

Assistant Professor of Biology, Grinnell College., Grinnell, IA — Aug 2026 - Present
Teach undergraduate courses in pharmacology, biochemistry, and computational biology in a one-year, non-tenure-track appointment. Responsibilities include course design, student mentoring, and integrating molecular, biochemical, and computational approaches into biology education.

Health AI Research Coop, EAI at Northeastern Univ., Portland, ME — Jan 2026 - Present
Led research integrating medical imaging data with bulk RNA-sequencing to study the role of senescent fibroblast populations in basal cell carcinoma. Responsibilities include data curation, computational analysis, and collaboration with computer vision and biomedical researchers.

Postdoctoral Fellow, Merck Research Laboratories, Boston, MA — Sept 2020 - May 2023
Conducted and managed an experimental and computational research program on fibroblast biology in pancreatic ductal adenocarcinoma. Performed analysis of bulk and single-cell RNA-sequencing data generated in-house to interpret senescent tumor microenvironment states. Collaborated with biologists and bioinformaticians to translate transcriptomic findings into actionable insights.

Intern (rotation), Merck Research Laboratories Ventures, Boston, MA — Oct 2021 - Jul 2022
The MRL Ventures Fund (MRLV) team invests in early-stage, preclinical therapeutics companies. This associate level internship position includes meeting with venture creators, early diligence, landscape analysis, investing, and large picture strategy.

Teaching Experience

Reflective Teaching Seminar, Brown University's Sheridan Center — Jan 2016 - May 2016
Completed the certificate program in inclusive, evidence-based pedagogy focused on course design, student engagement, assessment, and reflective teaching practice.

Portuguese I and II Instructor and Curriculum Dev., Grinnell College — Jan 2013 - May 2014
Developed and taught 1-credit Portuguese I and II courses; designed syllabus and assessments; 100%

student pass rate on external proficiency evaluation.
Biochemistry Teaching Assistant, Grinnell College — Aug - Dec 2013
General Chemistry Laboratory Assistant, Grinnell College — Jan - May 2013

Publications

Scaffa A, Hornburg M, Kopsiaftis S. Single-Cell Transcriptomics of Senescent Cancer-Associated Fibroblasts. 2024; [unpublished - industry research].

Yao H, Wallace J, Peterson A, **Scaffa A**, Rizal S, Hegarty K, Maeda H, Chang J, Oulhen N, Kreiling J, Huntington K, De Paepe M, Barbosa G, Dennery PA. Timing and cell specificity of senescence drives postnatal lung development and injury. Nature Commun 14, 273. 2023.

Scaffa A, Tollefson G, Yao, H, Rizal S, Wallace J, Oulhen N, Carr J, Hegarty K, Uzun A, Dennery PA. Identification of Heme Oxygenase-1 as a Putative DNA-Binding Protein. Antioxidants. 2022 Nov; 11:2135. DOI: 10.3390/antiox11112135.

Scaffa A, Yao, H, Oulhen, N, Dennery, PA. Single-cell transcriptomics reveals lasting changes in the lung cellular landscape into adulthood after neonatal hyperoxic exposure. Redox Biology. 2021 Dec; 48:102091. PMID: 34417156.

Scaffa AM, Peterson AL, Carr JF, Garcia D, Yao H, Dennery PA. Hyperoxia causes senescence and increases glycolysis in [...] epithelial cells. Physiol Rep. 2021 May; 9(10):e14839. PMID: 34042288.

Garcia D, Carr JF, Chan F, Peterson AL, Ellis KA, **Scaffa A**, Ghio AJ, Yao H, Dennery PA. Short exposure to hyperoxia causes cultured lung epithelial cell mitochondrial dysregulation and alveolar simplification in mice. Pediatr Res. 2021 Jul; 90(1):58-65. PMID: 33144707.

Carr JF, Garcia D, **Scaffa A**, Peterson AL, Ghio AJ, Dennery PA. Heme Oxygenase-1 Supports Mitochondrial Energy Production and Electron Transport Chain Activity in Cultured Lung Epithelial Cells. Int J Mol Sci. 2020 Sep; 21(18):6941. PMID: 32971746.

Scaffa A, Peterson, A, Dennery PA. Hyperoxic Exposure Increases Senescence [...] Role Of p53 Signaling. Am J Respir Crit Care Med. A61 Epithelial Biology. 2019; 199:A7327. doi:10.1164.

Scaffa A, Peterson, A, Dennery PA. Hyperoxic exposure causes senescence in [...] lung epithelial cells. Pediatric Academic Society. 3700 Pulmonology: Biology, Epidem., and Phys. 2019; 3700.6.

Skills

Computational & Analytical: Transcriptomic data analysis (bulk RNA-seq, scRNA-seq); statistical analysis and data visualization; reproducible workflows; applied ML for biological data.
Programming Languages: Python (pandas, numpy, sklearn, matplotlib, seaborn, xgboost, joblib); Bash; Java, R (basic); C/C++ (basic); HTML/CSS; React; Tailwind CSS.
Biological & Domain Expertise: Cancer biology; cellular senescence; tumor microenvironment; pharmaceutical R&D; CRISPR; metabolism; flow cytometry; project planning and management.
Languages: Fluent: English, Portuguese, and Spanish.

Computational Projects (Selected)

Machine Learning Pipeline for Cancer Classification, Northeastern Univ. — Jan - June 2025
Designed an end-to-end, reproducible ML pipeline for classification and prediction of cancer using public cancer RNA-sequencing data.
Markov-Model Sequence Generation Project, Northeastern Univ. — Aug - Dec 2024
Applied probabilistic modeling techniques commonly used in computational biology (Hidden Markov Models) to learn and generate structured sequences in a non-biological domain.

Honors and Awards

Tarr-Coyne Family Scholarship, The Roux Institute of Northeastern University — 2025
Bertelsmann Tech. Chall. Scholarship, Machine Learning, Bertelsmann and Udacity — 2021
Google Developer Challenge Scholarship, Google and Udacity — 2018
Best Aging in Place Hack, MIT Hacking Medicine Grand Hack, Cambridge, MA — 2016
Pharmacology Pre-Doctoral Fellow, Brown University, Providence, RI — 2015
1st Prize in Biochemistry, Best in Category, and naming rights of a minor planet — 2010
Intel® International Science and Engineering Fair (ISEF) co-hosted by Google, San Jose, CA

Community Involvement (Abbreviated)

Lead, Bioinformatics Interest Group, Northeastern University — Jan 2026 - Present
President of SACNAS, Brown University — Jul 2018 - Jun 2019
Member, Biomedical Science Career Program, Harvard Medical School — Jun 2016 - Present

Active member, panelist, abstract judge for NESS, and mentor to high school students.

CERTIFICATIONS

Python for Data Science and AI, Coursera, 2019. **Data Science Methodology**, Coursera, 2019. **Open Source Tools for Data Science**, Coursera, 2019. **A Crash Course in Data Science**, Coursera, 2018. **Teaching Certificate I**, Sheridan Center. Brown University, 2016.